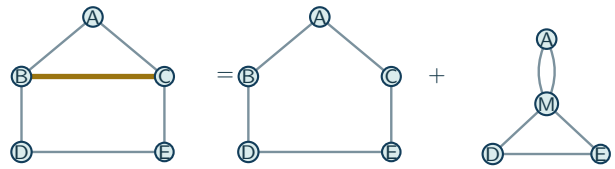
GRAPH THEORY LADDER 8 OF 9

The Laplacian and counting spanning trees

Degrees minus the adjacency matrix give a matrix whose determinant counts spanning trees.

1 / THE RULE AND THE PICTURE

Delete and contract an edge, and the counts add



house: 11 | delete BC: 5 | contract BC: 6

Spanning trees either avoid an edge e, in which case they live in $G - e$, or use it, in which case they correspond to trees of G/e with the ends of e merged, so

$$\tau(G) = \tau(G - e) + \tau(G/e)$$

The house gives $11 = 5 + 6$, the 5-cycle having 5 and the merged graph 6.

2 / THE LAPLACIAN MATRIX

Degrees on the diagonal and minus the adjacency matrix elsewhere

$$L = D - M = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 \\ -1 & -1 & 3 & 0 & -1 \\ 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

Every row sums to 0, the matrix is symmetric, and for any assignment of numbers x_v to the vertices $x^T L x = \sum_{uv \in E} (x_u - x_v)^2$, the energy of springs along the edges.
The number of zero eigenvalues equals the number of connected components, and for the house the eigenvalues are $0,\ 1.382,\ 2.382,\ 3.618,\ 4.618$.

3 / KIRCHHOFF'S MATRIX TREE THEOREM

Delete any row and the same column, and take the determinant

Removing row and column A leaves a 4×4 matrix whose determinant is 11, and removing any other vertex gives 11 again, so the house has 11 spanning trees.
Equivalently the product of the non-zero eigenvalues divided by n is $\frac{1.382 \times 2.382 \times 3.618 \times 4.618}{5} = 11$. The same recipe gives 16 for K_4, 125 for K_5 and 81 for $K_{3,3}$, in agreement with Cayley's formula and its bipartite version.

4 / KNOWN COUNTS

Standard graphs

graph	spanning trees	value
path P_n	1	nothing to choose
cycle C_n	n	leave out any one edge
complete graph K_n	n^{n-2}	K_4: 16, K_5: 125
complete bipartite $K_{m,n}$	$m^{n-1} n^{m-1}$	$K_{3,3}$: 81
house graph	11	the determinant below

5 / TRY THESE

Count by deleting and contracting

(a) A triangle has 3 spanning trees: delete one edge from three choices.
(b) The second smallest eigenvalue, 1.382 for the house, measures how well connected the graph is: it is 0 exactly when the graph is disconnected.
(c) Check $\tau(K_4) = 16$ by deleting an edge: $K_4 - e$ has 8 trees and K_4/e has 8.

ASIDE / ELECTRICAL NETWORKS

Kirchhoff introduced these matrices in 1847 to analyse electrical circuits, and the same determinant counts the trees that decide currents.

LINKS TO OTHER SHEETS

The adjacency matrix sheet defines M, and the earlier squared rectangles paper uses the same Kirchhoff flows.

Convention. $L = D - M$ with D the diagonal of degrees; $\tau(G)$ the number of spanning trees; $G - e$ deletes and G/e contracts the edge e.